

Integrated Diffractive Slab Photonic Processor for Real-Time RF Spectrum Sensing and Matched Filtering

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Abstract We demonstrate a silicon-nitride diffractive slab photonic processor for real-time RF spectrum sensing and matched filtering. A 24-point, 9.12-GHz one-shot processor achieves 380-MHz resolution and 3-ns-scale response, while frequency stitching improves matched-filtering resolution to 47.5 MHz and yields >20-dB sidelobe suppression in pulse compression. ©2026 The Author(s)

Introduction

Real-time high-speed radio-frequency (RF) systems for spectrum monitoring, radar, and electronic reconnaissance require repeated Fourier-transform and convolution-type operations under stringent latency and power constraints. As instantaneous bandwidth and update rate continue to increase, electronic implementations face growing pressure from processing throughput, data movement, and multi-stage latency [1, 2]. Integrated photonics provides an attractive hardware path for these structured operators [3], whereas many programmable coherent photonic processors achieve versatility at the cost of numerous tuning elements and electrical control channels [4].

Here, we report a diffractive slab photonic processor (DSPP) on a silicon nitride platform for real-time RF spectrum sensing and matched filtering. Compared with mesh-style coherent computing hardware, the DSPP targets structured operators with substantially reduced control complexity. In this paper, we focus on two system-level demonstrations: one-shot RF spectral estimation and radar-oriented matched filtering with stepped-frequency stitching.

Device concept and implementation

This DSPP chip is fabricated on a silicon nitride photonic platform using LIGENTEC's process. Its structure features an 800 nm thick silicon nitride layer and employs a 180 nm lithography process. As shown in Fig. 1(a), the complete DSPP chip occupies a compact footprint of 3.0 mm × 10.5 mm. The DSPP combines an on-chip serial-to-parallel front end with true-time-delay capability and a two-stage diffractive slab processing architecture. The first slab performs a 24-point discrete Fourier transform (DFT), while the second stage supports inverse transformation or system-function loading for structured processing [5, 6]. To enhance integration density, arrayed facet couplers are em-

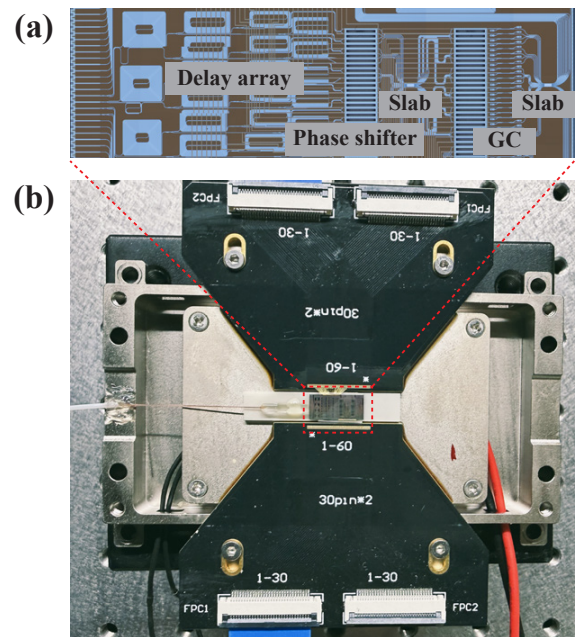


Fig. 1: (a) Microscope image of the fabricated DSPP chip. (b) Packaged module.

ployed, and electrical PAD-based packaging techniques are applied, resulting in the finalized chip presented in Fig. 1(b). In this work, we mainly exploit the front-end serial-to-parallel conversion and first-stage slab FFT for spectrum sensing and dechirped matched-filter readout.

The proposed pulse-compression receiver performs matched filtering and spectrum sensing directly in the optical domain, and the experimental setup is shown in Fig. 2. The two cascaded IQ modulators in the figure are used for the matched-filtering experiment, whereas only one modulator is required for spectrum sensing. A reference signal generated by a waveform generator is first split by a 90° hybrid to produce the in-phase (I) and quadrature (Q) electrical drive signals, which are then applied to the I and Q arms of the first IQ modulator operated in the carrier-suppressed +1st-order single-sideband state. The echo signal

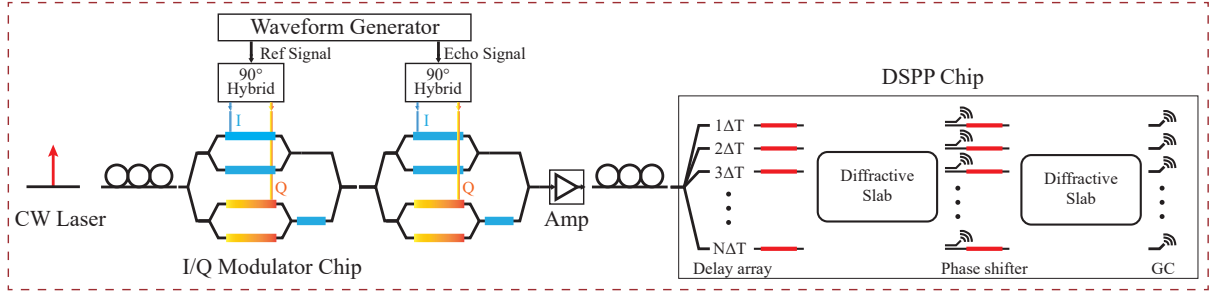


Fig. 2: Experimental setup.

is then applied to a second cascaded IQ modulator biased in the carrier-suppressed -1 st-order single-sideband state. Under this complementary sideband configuration, the reference and echo terms are combined within the optical field, so that the delay-dependent beat term is generated and can be directly resolved by the subsequent DSPP chip.

Starting from the optical carrier $E_0 e^{j\omega_0 t}$, the first modulator generates

$$E_1(t) = \alpha E_0 e^{j\omega_0 t} e^{j\phi_r(t)}, \quad (1)$$

where $\phi_r(t) = 2\pi f_s t + \pi K t^2$ is the phase of the reference linear frequency modulated (LFM) signal. The delayed echo has phase $\phi_e(t) = 2\pi f_s(t - \tau) + \pi K(t - \tau)^2$, and the second modulator contributes its conjugated term, yielding

$$E_2(t) = \alpha \beta E_0 e^{j\omega_0 t} e^{j[\phi_r(t) - \phi_e(t)]}. \quad (2)$$

So the output optical field contains a single-tone dechirped component,

$$E_2(t) \propto e^{j\omega_0 t} e^{j2\pi K \tau t}, \quad (3)$$

with beat frequency $f_b = K\tau$. Compared with conventional photonic dechirp receivers based on direct beating of independently modulated optical components at the photodetector, this cascaded-IQ scheme suppresses redundant carrier- and image-related beat products, directs more modulation energy to the useful delay-dependent term, and therefore improves the spectral contrast of the compressed pulse response. To further improve the effective frequency resolution, a small frequency shift was applied to the reference signal over repeated measurements, and the resulting spectra were stitched to form a finer equivalent FFT spectrum. For real-time spectrum sensing, only the first-stage IQ modulation in Fig. 2 is needed. By directly modulating the echo signal received by the antenna onto the optical carrier, real-time streaming detection of signals within the F_s band can be achieved.

Real-time spectrum sensing

In the experiment, real-time spectrum sensing is emulated by intensity-modulating a continuous-wave optical carrier with RF signals generated by

a waveform generator and launching the modulated light into the DSPP. By reading out the optical power from the 24 output ports in parallel, the DSPP directly provides frame-based spectral energy estimates without frequency scanning.

For a LFM signal $s(t) = \exp(j\pi K t^2)$, the phase variation within the FT window $[-T/2, T/2]$ is $\Delta\phi = \pi K T^2$. When $\Delta\phi > \pi$, defocusing blur occurs. In this experiment, the corresponding critical chirp rate is $K_{\max} \approx 1/T^2 = 1.44 \times 10^{17} \text{ Hz/s}$. Fig. 3(a) presents the response of the DSPP to a rapidly varying triangular LFM signal with a chirp rate of $1.4 \times 10^{17} \text{ Hz/s}$. The result clearly verifies the capability of the DSPP to track fast swept-frequency signals in real time. Fig. 3(b) further evaluates the device under a more extreme fast frequency-hopping condition, where the input periodically switches from 760 MHz to 5.32 GHz, 3.8 GHz, 1.9 GHz, 3.42 GHz, and 2.28 GHz, before hopping back to 0.76 GHz, with the hopping interval reduced to 3 ns. Notably, distinct spectral peaks remain observable in successive frames and the applied 3 ns hopping interval is close to the minimum processing window, providing experimental validation of the theoretical response limit.

As summarized in Table 1, the DSPP chip offers a significant advantage in temporal resolution over

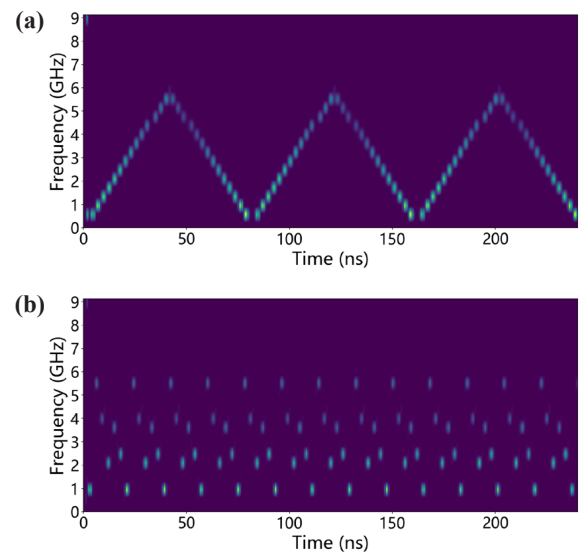


Fig. 3: (a) Processing of a high-speed LFM signal. (b) Processing results for a high-speed frequency-hopping signal

Tab. 1: Comparison of real-time RF spectrum sensing systems

Work	BW	Freq. Res.	Time Res.	Readout
Wang, PR 2019 [7]	33 GHz	15 MHz	~ 1 ms	Scanning
Zhang, OE 2023 [8]	29 GHz	375 MHz	~ 10 ms	Scanning
He, Sci. Adv. 2025 [9]	50.8 GHz	20 MHz	500 ns	Off-chip
Tao, NC 2026 [1]	57.5 GHz	350 MHz	100 ns	Channelized
Our slab-FFT	9.12 GHz	380 MHz	2.63 ns*	Parallel FFT

*One 24-point processing window at an effective sampling rate of 9.12 GHz.

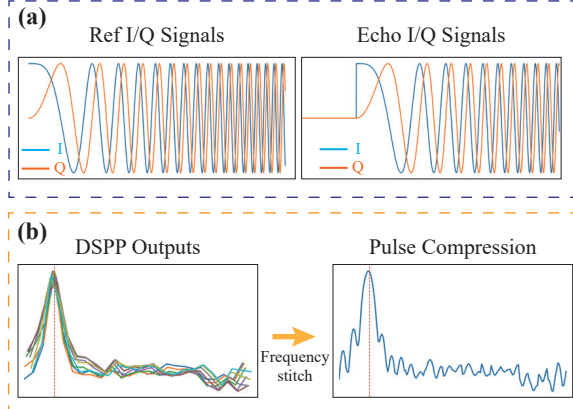


Fig. 4: (a) I/Q channels for the reference and echo signals. (b) DSPP outputs at different shift frequencies and the stitched high-resolution result.

representative real-time RF spectrum-sensing approaches. In particular, the nanosecond-scale response achieved here is substantially faster than the millisecond-scale response of electronic scanning systems and also exceeds other reported optical implementations with time resolutions on the order of hundreds of nanoseconds.

Radar-oriented Matched filtering

We further evaluate the processor in a radar-oriented matched-filtering experiment. A base-band LFM reference waveform is first modulated onto the optical carrier by an IQ modulator operated in the carrier-suppressed +1st-order single-sideband state, while the echo signal is applied to a second cascaded IQ modulator biased in the carrier-suppressed -1st-order single-sideband state, as shown in the left and right subplots of Fig. 4(a). The reference signal is still chosen as an LFM waveform with a chirp rate of $K = 1.4 \times 10^{17}$ and a bandwidth of $B = 5.6$ GHz. For the echo signal, an additional delay of approximately 5 ns is introduced, corresponding to a dechirped beat frequency of 0.7 GHz. We perform 8 repeated measurements, where in each reuse a single-tone frequency offset of $f = n \times 47.5$ MHz is added to the reference signal. By stitching the 24-channel output spectra obtained from the 8 repeated measurements, the effective spectral resolution is improved to 47.5 MHz. As clearly shown in Fig. 4(b), frequency stitching leads to a smoother matched-filtering peak and a sidelobe suppression ratio exceeding 20 dB.

Table 1 summarizes representative real-time RF spectrum sensing systems. Although the current demonstration targets a smaller instantaneous bandwidth, the DSPP offers a compact one-shot parallel architecture with nanosecond-scale temporal resolution and reduced hardware complexity.

Discussion

The current DSPP chip is mainly constrained by two factors: the on-chip serial-to-parallel conversion unit, whose performance is determined by the absolute waveguide length and the group index at the operating wavelength and thus sets the photonic FFT bandwidth, and the channel number of the diffraction unit, which determines the achievable spectral granularity within this bandwidth. The spectral granularity can be further improved by introducing a stepwise single-tone input sequence, which is approximately equivalent to synthesizing a larger-channel-count slab through spectral stitching. For spectrum sensing applications, the chip can complement conventional low-speed and slow-response electronic spectrum sensing schemes by enabling fast front-end preprocessing, while in radar imaging applications, carrier-suppressed single-sideband modulation implemented with an IQ modulator can significantly improve energy utilization efficiency, and the dechirped output can be directly used for subsequent two-dimensional imaging.

Conclusions

We have demonstrated an integrated silicon-nitride diffractive slab photonic processor for real-time RF spectrum sensing and radar-oriented matched filtering. The 24-point DSPP provides one-shot parallel spectral analysis over a 9.12 GHz bandwidth with 380 MHz resolution, and experimentally tracks rapidly varying LFM and frequency-hopping signals at nanosecond-scale update rates. By combining cascaded carrier-suppressed single-sideband IQ modulation with frequency stitching, the processor also performs optical-domain matched filtering with an effective spectral resolution of 47.5 MHz and a sidelobe suppression ratio exceeding 20 dB. These results establish the DSPP as a compact and low-complexity photonic hardware primitive for high-speed front-end RF signal processing.

Acknowledgements

This work was supported in part by the Key Program of Chinese Academy of Sciences under Grant RCJJ-145-24-16, and in part by the National Natural Science Foundation of China under Grant 62327806 and Grant U2441246.

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